

QUIZ 2(A1)

Name: _____

Roll# _____

Q1: For $f(x) = x^{\frac{4}{3}}(x - 2)$ answer each of the following questions..

- Determine the intervals on which the function is concave up and concave down.
- Determine the inflection points of the function.

Sol:

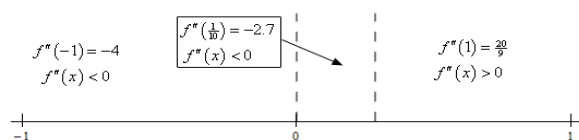
We'll need the 2nd derivative to find the list of possible inflection points.

$$f''(x) = \frac{28}{9}x^{\frac{1}{3}} - \frac{8}{9}x^{-\frac{2}{3}} = \frac{28x - 8}{9x^{\frac{2}{3}}}$$

The possible inflection points for this function are,

$$x = 0 \quad \& \quad x = \frac{2}{7} = 0.2857$$

To get the intervals of concavity we'll need the number line for the 2nd derivative.



From this we get the following concavity information for the function.

$$\text{Concave Up : } \left(\frac{2}{7}, \infty\right) \quad \text{Concave Down : } (-\infty, 0) \quad \& \quad \left(0, \frac{2}{7}\right)$$

From the concavity information in the previous step we can see that the single inflection point for the function is,

$$x = \frac{2}{7}$$

Q2: Q1: Use L'Hospital's Rule to evaluate $\lim_{x \rightarrow -\infty} x \cdot e^x$

So, it's in the form $(\infty)(0)$. This means that we'll need to write it as a quotient. Moving the x to the denominator worked in the previous example so let's try that with this problem as well.

$$\lim_{x \rightarrow -\infty} x e^x = \lim_{x \rightarrow -\infty} \frac{e^x}{1/x}$$

Writing the product in this way gives us a product that has the form $0/0$ in the limit. So, let's use L'Hospital's Rule on the quotient.

$$\lim_{x \rightarrow -\infty} x e^x = \lim_{x \rightarrow -\infty} \frac{e^x}{1/x} = \lim_{x \rightarrow -\infty} \frac{e^x}{-1/x^2} = \lim_{x \rightarrow -\infty} \frac{e^x}{2/x^3} = \lim_{x \rightarrow -\infty} \frac{e^x}{-6/x^4} = \dots$$

Hummm... This doesn't seem to be getting us anywhere. With each application of L'Hospital's Rule we just end up with another $0/0$ indeterminate form and in fact the derivatives seem to be getting worse and worse. Also note that if we simplified the quotient back into a product we would just end up with either $(\infty)(0)$ or $(-\infty)(0)$ and so that won't do us any good.

This does not mean however that the limit can't be done. It just means that we moved the wrong function to the denominator. Let's move the exponential function instead.

$$\lim_{x \rightarrow -\infty} x e^x = \lim_{x \rightarrow -\infty} \frac{x}{1/e^x} = \lim_{x \rightarrow -\infty} \frac{x}{e^{-x}}$$

Note that we used the fact that,

$$\frac{1}{e^x} = e^{-x}$$

to simplify the quotient up a little. This will help us when it comes time to take some derivatives. The quotient is now an indeterminate form of $-\infty/\infty$ and using L'Hospital's Rule gives,

$$\lim_{x \rightarrow -\infty} x e^x = \lim_{x \rightarrow -\infty} \frac{x}{e^{-x}} = \lim_{x \rightarrow -\infty} \frac{1}{-e^{-x}} = 0$$

QUIZ 2(A2)

Name: _____

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Q1: Find and classify all the critical points of the following function. Give the intervals where the function is increasing and decreasing. $f(t) = t \cdot \sqrt[3]{t^2 - 4}$.

First, we'll need the derivative so we can get our hands on the critical points. Note as well that we'll do some simplification on the derivative to help us find the critical points.

$$\begin{aligned} g'(t) &= (t^2 - 4)^{\frac{1}{3}} + \frac{2}{3}t^2(t^2 - 4)^{-\frac{2}{3}} \\ &= (t^2 - 4)^{\frac{1}{3}} + \frac{2t^2}{3(t^2 - 4)^{\frac{2}{3}}} \\ &= \frac{3(t^2 - 4) + 2t^2}{3(t^2 - 4)^{\frac{2}{3}}} \\ &= \frac{5t^2 - 12}{3(t^2 - 4)^{\frac{2}{3}}} \end{aligned}$$

So, it looks like we'll have four critical points here. They are,

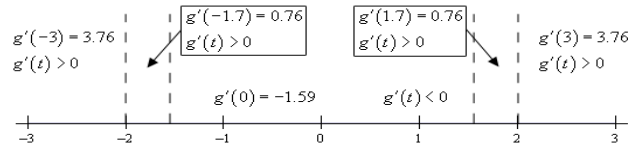
$$t = \pm 2$$

The derivative doesn't exist here.

$$t = \pm \sqrt{\frac{12}{5}} = \pm 1.549$$

The derivative is zero here.

Finding the intervals of increasing and decreasing will also give the classification of the critical points so let's get those first. Here is a number line with the critical points graphed and test points.



So, it looks like we've got the following intervals of increasing and decreasing.

$$\text{Increase : } -\infty < x < -2, \quad -2 < x < -\sqrt{\frac{12}{5}}, \quad \sqrt{\frac{12}{5}} < x < 2, \quad \& \quad 2 < x < \infty$$

$$\text{Decrease : } -\sqrt{\frac{12}{5}} < x < \sqrt{\frac{12}{5}}$$

From this it looks like $t = -2$ and $t = 2$ are neither relative minimum or relative maximums since the function is increasing on both side of them. On the other hand, $t = -\sqrt{\frac{12}{5}}$ is a relative maximum and $t = \sqrt{\frac{12}{5}}$ is a relative minimum.

Q2: Use L'Hospital's Rule to evaluate $\lim_{x \rightarrow 0^+} [x \ln x]$

Note that we really do need to do the right-hand limit here. We know that the natural logarithm is only defined for positive x and so this is the only limit that makes any sense.

Now, in the limit, we get the indeterminate form $(0)(-\infty)$. L'Hospital's Rule won't work on products, it only works on quotients. However, we can turn this into a fraction if we rewrite things a little.

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x}$$

The function is the same, just rewritten, and the limit is now in the form $-\infty/\infty$ and we can now use L'Hospital's Rule.

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2}$$

Now, this is a mess, but it cleans up nicely.

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0$$